

Throws App Functionality and Workflow Summary

1. App Identity

This app is a **self-hosted throws training operating system** for managing track and field throwing athletes.

It is not just a workout log. It is designed to help the coach:

- Plan practices faster
- Assign training more precisely
- Track thrower-specific data
- Monitor practice live
- Catch PRs and bad trends
- Adjust training based on evidence
- Build a long-term performance history for each athlete

The app will be built as a **web app**, accessed through a browser, with a **Python backend** and an **HTML / JavaScript / CSS frontend**.

2. Core Users

Coach

The coach has full control of the system.

The coach can:

- Create athletes
- Create groups
- Build training modules
- Create practice plans

- Assign workouts
- Edit individual athlete plans
- Monitor live practice progress
- Review throwing data
- Review lifting data
- Review meet results
- Track PRs
- Add notes
- View weekly reports
- Approve projected max changes

Athlete

The athlete uses the app to follow the plan and enter data.

The athlete can:

- View today's Practice plan
- view today's Lifting plan
- Check off completed work
- Log throws
- Log lifting results
- Enter meet performance
- View PRs
- View weekly reports
- Complete wellness check-ins if assigned

Guardian

Guardian access is planned for later but is not required for the first version.

3. Main Functional Areas

3.1 Athlete Management

Each athlete profile should include:

- Name
- Sex
- Event group or groups(Shotput, Discus, Javelin)
- Event-specific PRs
- Implement-specific PRs (Light, Medium, Heavy)
- Weight room projected maxes
- Meet history
- Coach notes
- Weekly report history

Athletes can belong to multiple event groups, such as shot put and discus.

3.2 Calendar-Based Planning

The coach uses a calendar to organize training.

Each calendar day can include:

- Practice plan
- Weight room plan
- Meet day (if Applicable)
- Custom Events
- Notes
- Reports

The calendar should make it easy to look back and answer:

- What did we do?
- Who completed it?
- What marks were recorded?

- Did anyone PR?
- Did the training appear to work?

3.3 Reusable Training Modules

The app is built around reusable training modules.

the coach is responsible for filling out the meet library, needs to allow for future plans to be added

Each module should contain: Module Name, Variation, Reps, Measured or Non Measured (if applicable), notes.

each module should have a show more option edited by the coach to allow for images or short videos for demonstration purposes and also a section for "coaching cues" for the coach to add specific coaching points for that specific module or "info" to leave room for explanation of the module. This should be viewable on the athlete side as well.

Example modules:

- Daily warmup
- Ankle walks
- Hurdle mobility
- ect

The coach can create a module once and reuse it across many practices.

3.4 Practice Plan Builder

The coach can build a daily practice plan by selecting modules and adding specific assignments.

The coach can:

- Open a calendar date
- Add modules

- Assign the plan to a group
- Edit plans for individual athletes
- Publish the plan

The builder should work best on laptop or tablet but remain usable on a phone.

3.5 Athlete Practice View

Athletes open the web app in their phone browser and see today's plan.

The athlete practice view should show:

- Today's practice title
- Ordered practice sections
- Drill names
- Reps, throws, or time
- Implement weight or category
- Technical cues
- Checkoff buttons
- Data entry prompts when needed (measured throws)

For the first version, a task is complete when the athlete checks it off.

3.6 Throw Tracking

The app supports two throw tracking modes.

Standard Practice Mode

Used for normal practices.

Athlete enters:

- Low mark
- Typical mark

- Best mark
- Optional note

Specific Tracking Session Mode

Used when the coach wants detailed data.

Athlete enters every throw:

- Throw number
- Event
- Implement
- Throw type
- Distance
- Optional foul/miss flag
- Optional note

Athletes enter marks in feet and inches. The app automatically displays meters as a smaller secondary value.

3.7 Implement and PR Tracking

The app tracks PRs by:

- Athlete
- Event
- Implement weight
- Implement category
- Throw type

Shot Put PR Categories

- Stand throw
- Wheel / half technique
- Full throw

Discus PR Categories

- Stand throw
- Wheel / half technique
- Full throw

Javelin PR Categories

- 3-step
- 5-step
- Full approach

Implement Labels

The app uses the coach's existing language:

- Extra light
- Light
- Competition
- Heavy
- Extra heavy
- Custom

3.8 Alerts and Performance Flags

The app flags important throwing results based on the athlete's PR for the same event, implement, and throw type.

Initial rules:

- Good throw: 95% or better of PR
- Subpar throw: under 70% of PR
- Prolonged underperformance: repeated marks below 85%
- Bad trend: 4–6 poor throws in a row
- Intervention flag: 2–3 practices in the same event with no clear success or improvement

The coach remains in control of reviewing important alerts.

3.9 Coach Live Dashboard

The coach live dashboard shows athlete progress during practice.

Each athlete tile should show:

- Athlete name
- Profile image or initials
- Current practice section
- Completion percentage
- Current best mark
- Alert status
- Short status blurb

Example status blurbs:

- On track
- Strong session
- New PR pending review
- Below normal range
- Missing data
- Practice complete
- Needs review

For the web app MVP, the dashboard can update using JavaScript polling every 15–30 seconds.

3.10 Weight Room Tracking

The app supports a custom weight room system.

Main lifts:

- Bench
- Front squat
- Hang clean
- Hex bar deadlift
- Snatch

Initial block structure:

- Week 1: 10x2 @ 65–70%
- Week 2: 8x2 @ 75%
- Week 3: 5x2 @ 85–90%
- Week 4: 2x2 @ 95%, 1x1 @ 100–110%
- Week 5: deload / recovery module

The second set is treated as a burnout set.

Burnout performance generates a suggested projected max increase, but the coach must approve it before it affects future prescriptions.

3.11 Meet Performance Tab

Athletes can enter meet results.

Fields:

- Event
- Distance in feet/inches
- Automatic meter conversion
- Date
- Location
- Optional notes

For MVP, meet tracking is simple and does not require full attempt-by-attempt series tracking. a toggle for full event tracking (3-6 throws) should be available but only the best mark is needed.

3.12 Weekly Reports

Weekly reports summarize the training week.

Reports should include:

- Best practice of the week
- Best marks
- Worst marks
- Coach notes
- Trend line graph
- Practice completion summary
- PRs or near-PRs
- Underperformance flags

The report should help the athlete understand their week and help the coach decide whether training needs to change.

3.14 Notes and Comments

Coach can attach notes to:

- Athlete
- Practice day
- Throw result
- Lift result
- Weekly report

A weekly coach-notes summary should collect all notes from the week.

4. Overall Workflow

Step 1: Coach Sets Up the System

The coach creates the foundation:

- Athlete accounts
- Athlete profiles
- Event groups
- Implement lists
- PR baselines
- Training groups
- Drill/module library

This setup allows the app to assign training and track performance correctly.

Step 2: Coach Builds Reusable Modules

The coach turns normal practice pieces into reusable modules, these are examples :

- Warmup
- Ankle walks
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- Weight room blocks
- Recovery modules

This prevents the coach from rebuilding the same practice pieces every day.

Step 3: Coach Creates a Daily Plan

The coach opens the calendar and selects a date.

The coach builds the plan by adding modules and assignments.

Example daily structure:

1. Warmup
2. Ankle walks
3. Event-specific warmup
4. Drill block
5. Throwing assignment
6. Weight room or recovery work
7. Notes and cues

The coach can assign the plan to a group and then modify individual athletes as needed.

Step 4: Athlete Opens Today's Plan

The athlete logs into the web app through a browser.

The athlete sees only what they need for the day:

- What to do
- How much to do
- What implement to use
- What cue to focus on
- What data to enter

The athlete view should be fast, simple, and easy to use during practice.

Step 5: Athlete Completes and Logs Work

The athlete checks off tasks as they complete them.

When required, the athlete enters:

- Throw data
- Lift data
- Burnout reps

- Meet performance
- Notes

Standard practice logging should stay quick. Detailed tracking should only happen when the coach requests it.

Step 6: Coach Monitors the Live Dashboard

During practice, the coach opens the live dashboard.

The dashboard helps answer:

- Who is on track?
- Who is behind?
- Who hit a good mark?
- Who may have a PR?
- Who is underperforming?
- Who has missing data?

The dashboard is designed to make practice easier to manage in real time.

Step 7: App Flags Important Results

The app checks logged data against PRs and expected ranges.

It can flag:

- New PR
- Near PR
- Good throw
- Subpar result
- Bad trend
- Suggested weight room max increase
- Missing or incomplete data

Coach remains in control of approving important changes.

Step 8: Coach Reviews and Adjusts

After practice, the coach reviews:

- Completion
- Throw marks
- PR flags
- Athlete notes
- Weight room results
- Burnout suggestions
- Meet entries

The coach uses the information to adjust future practices.

Step 9: Weekly Report Is Generated

At the end of the week, the app summarizes:

- Best marks
- Worst marks
- Best practice
- Trend line
- Coach notes
- PRs
- Performance flags

This gives both the coach and athlete a clear view of whether the week's training worked.

5. Simplified Summary

The app workflow is:

Plan → Assign → Complete → Log → Monitor → Flag → Review → Adjust → Report

The app should make it easier for the coach to run meaningful practices and make decisions from actual athlete data.